Solve with us 8 - Not Graded

This assignment will not be graded and is only for practice.

Key Points: 1Key Points: 1

- Two non-zero vectors uu and vv from an inner product space VV are said to be orthogonal if $\langle u, v\rangle = 0.\langle$u, v$\rangle = 0$.
- A set SS of non-zero vectors is said to be an orthogonal set of an inner product space if elements of the set SS are mutually orthogonal. i.e., $\langle v_i, v_j\rangle = 0\langle$vi, vj$\rangle = 0$ for $i \neq j$i = j, any $v_i, v_j \in S$vi, vj ∈ S.
- Let $\{v_1, v_2, \ldots, v_n\}${v1, v2, … , vn} be an orthogonal set of vectors in an inner product space. Then $\{v_1, v_2, \ldots, v_n\}${v1, v2, … , vn} is a linearly independent set of vectors.
- Orthogonal basis: Let VV be an inner product space. A basis consisting of mutually orthogonal vectors is called an orthogonal basis. In other word, a maximal orthogonal set is a basis and is called an orthogonal basis.
- An orthonormal set of vectors of an inner product space VV is an orthogonal set of vectors such that the norm of each vector of the set is 1.
- An orthonormal basis is an orthonormal set of vectors which form a basis.
- Obtaining orthonormal set from orthogonal set:Obtaining orthonormal set from orthogonal set: If $\gamma = \{v_1, v_2, \ldots, v_n\}$γ = {v1, v2, … , vn} is an orthogonal set of vectors then we can obtain an orthonormal set of vectors ββ from γγ by dividing each vector by its norm i.e.,
$$\beta = \left\{\frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}, \ldots, \frac{v_n}{\|v_n\|}\right\}.\beta = \{\|$v1$\|$v1, $\|$v2$\|$v2, … , $\|$vn$\|$vn} .$$

Consider a set of vectors
$S = \{(1, -1, -1), (1, 2, 0), (-1, 1, 1), (-1, \frac{1}{2}, -\frac{3}{2}), (0, 3, 1), (2, 1, 1), (0, 1, -1)\}$S = {(1, −1, −1), (1, 2, 0), (−1, 1, 1), (−1, 21, −23), (0, 3, 1), (2, 1, 1), (0, 1, −1)} from the inner product space R^3 R3 with usual inner product (i.e., the dot product).
Answer questions (1) and (2).

1 point
Which of the following subsets of the set SS consist of mutually orthogonal vectors?

☐ $\{(1, -1, -1), (1, 2, 0), (-1, 1, 1)\}${(1, −1, −1), (1, 2, 0), (−1, 1, 1)}

☐ $\{(1, -1, -1), (-1, \frac{1}{2}, -\frac{3}{2})\}${(1, −1, −1), (−1, 21, −23)}

☐ $\{(1, -1, -1), (-1, \frac{1}{2}, -\frac{3}{2}), (1, 2, 0)\}${(1, −1, −1), (−1, 21, −23), (1, 2, 0)}

☐ SS

☐ $\{(2, 1, 1), (1, -1, -1), (0, 1, -1)\}${(2, 1, 1), (1, −1, −1), (0, 1, −1)}

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(1, -1, -1), (-1, \frac{1}{2}, -\frac{3}{2})\}${(1, −1, −1), (−1, 21, −23)}
$\{(2, 1, 1), (1, -1, -1), (0, 1, -1)\}${(2, 1, 1), (1, −1, −1), (0, 1, −1)}

1 point
Which of the following option(s) is (are) true?

☐ $\{(2,1,1),(0,1,-1)\}\{(2,1,1),(0,1,-1)\}$ is an orthogonal basis of a vector space which is isomorphic to R^2 R2.

☐ $\{(2,1,1)\}\{(2,1,1)\}$ is not an orthogonal basis of a vector space which is isomorphic to RR.

☐ $\{(1,-1,-1),(1,2,0)\}\{(1,-1,-1),(1,2,0)\}$ is an orthogonal basis of a vector space which is isomorphic to R^2 R2.

☐ $\{(-1,1,1),(-1,\frac{1}{2},-\frac{3}{2}),(0,3,1),\}\{(-1,1,1),(-1,21,-23),(0,3,1),\}$ is an orthogonal basis of R^3 R3.

☐ $\{\frac{(2,1,1)}{\sqrt{6}},\frac{(1,-1,-1)}{\sqrt{3}},\frac{(0,1,-1)}{\sqrt{2}}\}\{\quad 6$
$\sqrt{}$
$(2,1,1),\quad 3$
$\sqrt{}$
$(1,-1,-1),\quad 2$
$\sqrt{}$
$(0,1,-1)\}$ is an orthonormal basis of R^3 R3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(2,1,1),(0,1,-1)\}\{(2,1,1),(0,1,-1)\}$ is an orthogonal basis of a vector space which is isomorphic to R^2 R2.

$\{\frac{(2,1,1)}{\sqrt{6}},\frac{(1,-1,-1)}{\sqrt{3}},\frac{(0,1,-1)}{\sqrt{2}}\}\{\quad 6$
$\sqrt{}$
$(2,1,1),\quad 3$
$\sqrt{}$
$(1,-1,-1),\quad 2$
$\sqrt{}$
$(0,1,-1)\}$ is an orthonormal basis of R^3 R3.

Key Points: 2Key Points: 2

- The projection of a vector to a subspace:The projection of a vector to a subspace: Let VV be an inner product space, $v \in V$v ∈ V and $W \subseteq V$W ⊆ V be a subspace. Then the projection of vv onto WW is the vector in WW, denoted by $proj_W(v)$projW (v), computed as follows:

  Find an orthonormal basis $\{v_1, v_2, \ldots, v_n\}$ for W. Find an orthonormal basis {v1, v2 , … , vn} for W.

  Then $proj_W(v) = \sum_{i=1}^{n} \langle v, v_i \rangle v_i$ .Then projW (v) = ∑i=1n⟨v, vi⟩vi.

  Remark: $proj_W(v)$projW (v) is independent of the chosen orthonormal basis.
- Let VV be an inner product space and WW be a subspace. Then the projection of vectors in VV to W W is a linear transformation from VV to VV with image WW.

1 point

Consider the vector space R^3 R3 with usual inner product. Which of the following vectors represents projection of the vector u = (2, 1, −2)u = (2, 1, −2) onto the subspace whose an orthonormal basis is

$\{(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}), (\frac{-2}{3}, \frac{1}{3}, \frac{2}{3})\}\{(\quad 2$
$\sqrt{}$
$1, 0, \quad 2$
$\sqrt{}$
$1), (3-2, 31, 32)\}?$

$(\frac{14}{9}, \frac{-7}{9}, \frac{-14}{9})(914, 9-7, 9-14)$
(1,1,1)
(2,1,-2)

$(\sqrt{2}, 0, \sqrt{2})(\quad 2$
$\sqrt{}$
$, 0, \quad 2$
$\sqrt{}$
$)$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(\frac{14}{9}, \frac{-7}{9}, \frac{-14}{9})(914, 9-7, 9-14)$

Consider the vector space R^3 R3 with usual inner product. If (a, b, c)(a, b, c) is the projection vector obtained from projecting the vector $(1, 0, 1)$(1, 0, 1) onto the subspace $\{(x, y, z) \mid x - y + z = 0, x, y, z \in R\}$\{(x, y, z) | x − y + z = 0, x, y, z ∈ R\}, then find the value $3(a + b + c)$3(a + b + c).

Hint:Hint: Find any orthonormal basis of the subspace $\{(x, y, z) \mid x - y + z = 0, x, y, z \in R\}$ \{(x, y, z) | x − y + z = 0, x, y, z ∈ R\} and use the definition of the projection of a vector onto a subspace.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 4
1 point

Key Points: 3Key Points: 3

- Gram-schmidt process:Gram-schmidt process: The Gram-schmidt process is used to get an orthogonal basis from any given basis of a vector space. A slight modification yields an orthonormal basis.
- Let $\gamma = \{v_1, v_2, \ldots, v_n\}$γ = {v1, v2, … , vn} be a given ordered basis of a vector space. The steps in the process are described below and yield an orthonormal basis $\beta = \{u_1, u_2, \ldots, u_n\}$β = {u1, u2 , … , un}.
  Step-1 $w_1 = v_1$ w1 = v1 , and $u_1 = \frac{w_1}{\|w_1\|}$u1 = ‖w1‖w1.
  Step-2 $w_2 = v_2 - \langle v_2, u_1 \rangle u_1$ w2 = v2 − ⟨v2, u1⟩u1 and $u_2 = \frac{w_2}{\|w_2\|}$u2 = ‖w2‖w2.

  Step-3 $w_3 = v_3 - \langle v_3, u_1 \rangle u_1 - \langle v_3, u_2 \rangle u_2$ w3 = v3 − ⟨v3, u1⟩u1 − ⟨v3, u2⟩u2 and $u_3 = \frac{w_3}{\|w_3\|}$u3 = ‖w3‖w3.

⋮ ⋮

Step-i $w_i = v_i - \sum_{j=1}^{i-1} \langle v_i, u_j \rangle u_j$ wi = vi − ∑j=1i−1⟨vi, uj⟩uj and $u_i = \frac{w_i}{\|w_i\|}$ui = ‖wi‖wi.

⋮ ⋮

Step-n $w_n = v_n - \sum_{j=1}^{n-1} \langle v_n, u_j \rangle u_j$ wn = vn − ∑j=1n−1⟨vn, uj⟩uj and $u_n = \frac{w_n}{\|w_n\|}$un = ‖wn‖wn.

Let ββ be an ordered basis of the inner product space R^3 R3 with usual inner product, where $\beta = \{(1, 1, -1), (1, 1, 0), (1, 0, 0)\}$β = {(1, 1, −1), (1, 1, 0), (1, 0, 0)} and $\gamma = \{u_1, u_2, u_3\}$γ = {u1, u2, u3} be the corresponding orthonormal basis obtained using the Gram-schmidt process. Use this information to answer the questions (5), (6) and (7).

If $u_1 = (a_1, b_1, c_1)$u1 = (a1, b1, c1), then find the value of $\sqrt{3}(a_1 + b_1 + c_1)$. 3

√

(a1 + b1 + c1).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

If $u_2 = (a_2, b_2, c_2)$u2 = (a2, b2, c2), then find the value of $\sqrt{6}(a_2 + b_2 + c_2)$. 6

√

(a2 + b2 + c2).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 4
1 point

If $u_3 = (a_3, b_3, c_3)$u3 = (a3, b3, c3), then find the value of $\frac{1}{\sqrt{2}}(a_3 + b_3 + c_3)$. 2

√

1(a3 + b3 + c3).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

Key Points: 6Key Points: 6

- Orthogonal transformation:Orthogonal transformation: Let VV be an inner product space and TT be a linear transformation from VV to VV. TT is said to be an orthogonal transformation if

$$\langle Tv, Tw \rangle = \langle v, w \rangle, \ \forall v, w \in V \langle \mathrm{T}v, \mathrm{T}w \rangle = \langle v, w \rangle, \ \forall v, w \in \mathrm{V}$$

- Orthogonal matrix:Orthogonal matrix: A square matrix AA is called an orthogonal matrix if $AA^T = A^T A = I$AAT = AT A = I(identity matrix)

1 point

Consider the following linear transformations $T_1, T_2 : R^2 \rightarrow R^2$, and $T_3, T_4 : R^3 \rightarrow R^2$ T1, T2 : R2 → R2, and T3, T4 : R3 → R2 defined as

$$T_1(x, y) = (\frac{x+y}{\sqrt{2}}, \frac{x-y}{\sqrt{2}}), \text{T1(x, y)} = (\quad 2$$

√

x+y, 2

√

x−y),

$$T_2(x, y) = (x + 3y, 2x - 2y), \text{T2(x, y)} = (\text{x + 3y, 2x − 2y}),$$

$$T_3(x, y, z) = (\frac{x+z}{\sqrt{2}}, \frac{x+y\sqrt{3}+z}{\sqrt{3}}), \text{T3(x, y, z)} = (\quad 2$$

√

x+z, 3

√

x+y 3

√

+z),

$$T_4(x, y, z) = (x + y + z, x + y), \text{T4(x, y, z)} = (\text{x + y + z, x + y}),$$

where R^3 R3 and R^2 R2 are the inner product spaces with respect to the dot product. Which of the linear transformation is an orthogonal transformation?

◯ T_1 T1

◯ T_2 T2

◯ T_3 T3

◯ T_4 T4

No, the answer is incorrect.

Score: 0

Accepted Answers:

T_1 T1

1 point

Which of the following option(s) is (are) true?

☐ $\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$ is an orthogonal matrix.

☐ $\frac{1}{\sqrt{2}}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}$ 2

√

1 [11−11] is an orthogonal matrix.

☐

$\begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ -\frac{1}{2} & 1 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$ is an orthogonal matrix

$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$ $\sqrt{2}$

$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$ $\sqrt{2}$ is an orthogonal matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}$ is an orthogonal matrix.

$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ $\sqrt{2}$

$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ $\sqrt{2}$ is an orthogonal matrix.

$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$ $\sqrt{2}$

$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$ $\sqrt{2}$ is an orthogonal matrix.